

Integration Map — Prior Capstone Projects → Clinical Triage System

This project integrates methods and design rationale from five prior capstone projects. Code is **rebuilt** for this submission to keep the artifact independently runnable; what is reused is *patterns, design discipline, and lessons learned*. Each module in [src/](#) carries a short header attributing its lineage.

#	Prior Project	Methods / Insights Adapted	Where in this system
P2	Data & Statistical Reasoning — UCI Bank Marketing, chi-square test of independence	Structured Initial Data Analysis before modeling; chi-square + Cramer's V for categorical association; explicit Limitations & Bias framing (selection bias, leakage caveats)	notebooks/01_data_exploration.ipynb ; paper §Industry Context & §Limitations
P3	Machine Learning — UCI Online Retail II, K-Means RFM segmentation	log1p + StandardScaler pipeline for skewed numeric features; scikit-learn Pipeline / ColumnTransformer ; reporting cluster/cohort sizes alongside metrics; PCA(2) for interpretable visualization	src/preprocessing.py , src/ml_model.py , integrated notebook visualization
P4	Deep Learning — Fashion-MNIST CNN with dropout regularization	PyTorch training loop with fixed seed; dropout; per-class / per-slice disaggregated evaluation (the "non-uniform error redistribution" insight); CPU-budget honesty in reporting	src/dl_model.py ; src/evaluation.py (bias audit by sex / age band)
P5	Generative AI — Convolutional VAE on CIFAR-10 vehicles	Responsible-AI framing for generative outputs: structural mitigations baked into design choices; "synthesis without provenance" risk; under-claiming model capability; no model-weight redistribution	src/genai_explainer.py guardrails; paper §Ethics (design-choice → mitigation table)
P6	Agentic AI — Research Brief Agent (plan + RAG + evaluator)	Plan → Route → Synthesise → Evaluate → Revise loop; specialised prompt-pattern agents; Chroma + OpenAI text-embedding-3-small RAG; tool registry; INSUFFICIENT EVIDENCE escape valve; refusal list;	src/rag/ , src/agent_orchestrator.py , outputs/run_log.jsonl

#	Prior Project	Methods / Insights Adapted	Where in this system
		runtime caps; sha256 ingest manifest; JSONL run log	

What is new in this project

- Industry context: clinical triage / cardiovascular risk decision support.
- Dataset: UCI Heart Disease (supervised binary classification on tabular data).
- Cross-domain integration: ML + DL + RAG + GenAI + agentic orchestration in one workflow.
- Healthcare-specific responsible-AI analysis: bias slices, hallucination mitigation via grounding, human-in-the-loop gating, "not for clinical use" disclaimers.

What is explicitly *not* reused

- No prior-project source code is copied or vendored.
- Datasets from prior projects (Bank Marketing, Online Retail II, Fashion-MNIST, CIFAR-10) are not used here.
- Prior model weights are not reused.